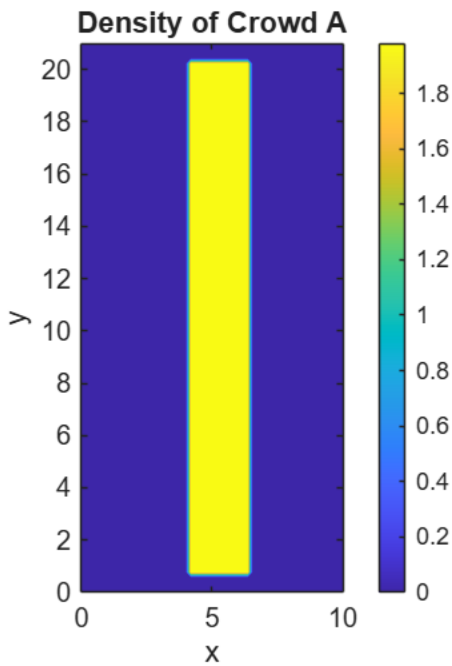
Mumbai Sub-Urban Local Train Bottleneck Problem

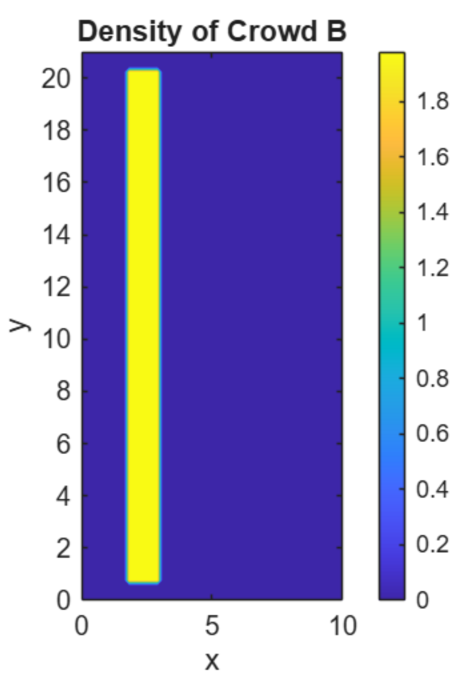
```
clear all;
clc;

%-------Domain Parameter-----%
% Defining velocity
global INIFNITE eta CFL speed_A speed_B discomfort_A discomfort_B cost_A cost_B
x_velocity_A y_velocity_A x_velocity_B y_velocity_B;

% Grid Points
global Nx; Nx = 50 + 6;
global Ny; Ny = 105 + 6;


%--Important Parameter
INFINITE = 10^12;
eta = 10^-6;
error_eta = 10^-9;
CFL = 0.5;

% To track the error for each iterations
iteration_error = [];

% Dimension of Domain
Lx = 10;
Ly = 21;

% Domain
x = linspace(0,Lx,Nx);
y = linspace(0,Ly,Ny);
[X,Y] = meshgrid(x,y);

% Space Interval
global h;
h = Lx/(Nx-6);

% Total Time (A typical Mumbai Sub-Urban Train halts for about 15 to 30
% seconds)
t_total = 10;

% --- Obstacle modeling ----%
global obstacle;
obstacle = false(Ny,Nx);
obstacle(1:18,19:20) = true; % Obstacle 1
% Region from Ny = 19 to Ny = 3 is the first region of inflow
obstacle(34:48,19:20) = true; % Obstacle 2
% Region from Ny = 49 to y = 63 is the second region of inflow
obstacle(64:78,19:20) = true; % Obstacle 3
% Region from Ny = 79 to Ny = 93 is the third region of inflow
```

```
obstacle(94:Ny,19:20) = true; % Obstacle 4

%-------Initial Condition------%
% Zero Density initial condition
rho_a = makeInitialDensity('A');  % Density of A
rho_b = makeInitialDensity('B');  % Density at B
rho_c = makeInitialDensity('C');  % Density at C

rho = rho_a + rho_b + rho_c; % Total Density at each point

%------Multiple Crowd Description------%
% Rho_max and V_max
rho_max_a = 10;
rho_max_b = 10;
v_max_a = 2;
v_max_b = 2;
% Discomfort
k_a = 0.002;
k_b = 0.002;

%--Function to compute Velocity using Density. Beware that this rho will be
% the total density in the domain.
function speed = makeSpeed(rho,v_max,rho_max)
    global Nx Ny;
    speed = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            speed(j,i) = v_max*(1 - rho(j,i)/rho_max);
        end
    end
end

%--Function to compute Discomfort using (rho here is the total density)
function discomfort = makeDiscomfort(rho,k)
    global Nx Ny;
    discomfort = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            discomfort(j,i) = k*rho(j,i)*rho(j,i);
        end
    end
end

%--Function to compute Cost function using Speed and Discomfort
function cost = makeCost(speed,discomfort)
    global Nx Ny;
    cost = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            cost(j,i) = 1/speed(j,i) + discomfort(j,i);
```

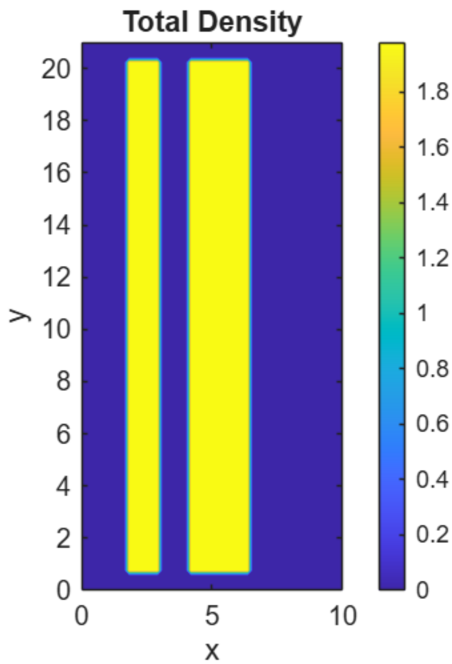
```
        end
    end
end
% Function that will calculate the flux depending on the u,v and Density.
% X_velocity and y_velocity will be found using the potential.
function flux = makeFlux(rho,x_velocity,y_velocity)
    global Nx Ny;

    % Initializing Flux
    flux_x = zeros(Ny,Nx);
    flux_y = zeros(Ny,Nx);
    flux = zeros(2, Ny, Nx);

    for i = 1:1:Nx
        for j = 1:1:Ny

            flux_x(j,i) = rho(j,i) * x_velocity(j,i);
            flux_y(j,i) = rho(j,i) * y_velocity(j,i);

            flux(1,j,i) = flux_x(j,i);
            flux(2,j,i) = flux_y(j,i);
        end
    end
end

%---Plots to verify the assignment

% Plotting the Density of the crowd A
figure(1);
contourf(X, Y, rho_a, 100, 'LineStyle', 'none');
colorbar;
title('Density of Crowd A');
xlabel('x');
ylabel('y');
axis equal tight;
```



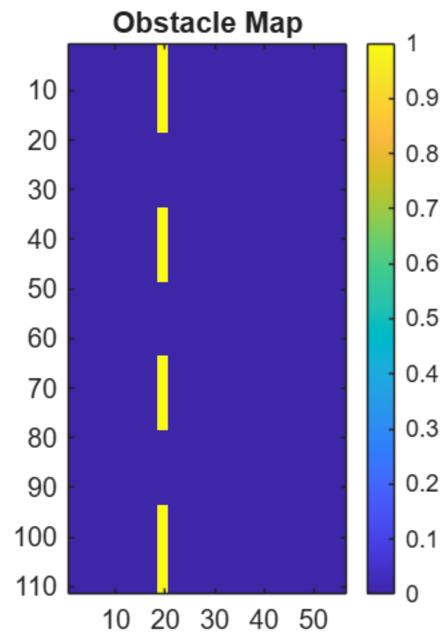

```
% Plotting the density of the crowd B
figure(2);
contourf(X, Y, rho_b, 100, 'LineStyle', 'none');
colorbar;
title('Density of Crowd B');
xlabel('x');
ylabel('y');
axis equal tight;
```



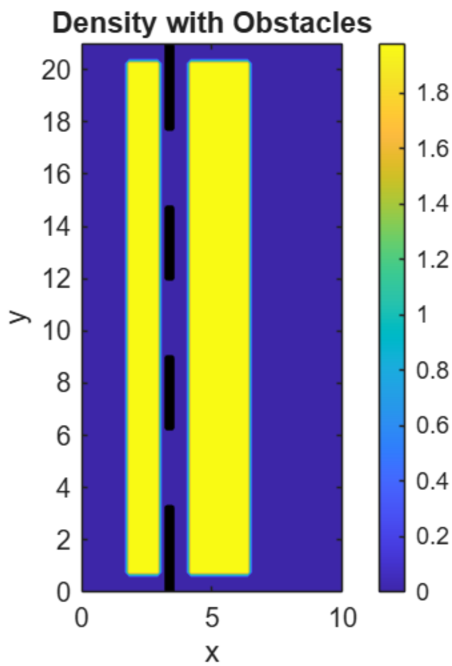
```
% Plotting the total density
figure(3);
contourf(X, Y, rho, 100, 'LineStyle', 'none');
colorbar;
title('Total Density');
xlabel('x');
ylabel('y');
axis equal tight;
```




```
% Plotting the obstacle
figure(4);
imagesc(obstacle);
colorbar;
title('Obstacle Map');
axis equal tight;
```




```
% Plotting the obstacle on density
figure(5);
contourf(X, Y, rho, 100, 'LineStyle', 'none');
hold on;
contour(X, Y, obstacle, [1 1], 'k', 'LineWidth', 2);
colorbar;
title('Density with Obstacles');
xlabel('x');
ylabel('y');
axis equal tight;
hold off;
```




```
%--Pre allocating the memory
rho_n_A = zeros(Ny,Nx);     rho_n_B = zeros(Ny,Nx);
rho_1_A = zeros(Ny,Nx);     rho_1_B = zeros(Ny,Nx);
rho_2_A = zeros(Ny,Nx);     rho_2_B = zeros(Ny,Nx);
rho_next_A = zeros(Ny,Nx);  rho_next_B = zeros(Ny,Nx);

% Stopping condition
stop_low = -1;
stop_high = 11;

% This will be used for plotting
x_plot = linspace(0, Lx, Nx-6);
y_plot = linspace(0, Ly, Ny-6);
[~, ix] = min(abs(x_plot - 36));    % index closest to x = 36
[~, iy] = min(abs(y_plot - 33));    % index closest to y = 33


%-----------MAIN SOLVER------------%
t = 0;
step = 0; % For plotting at regular interval

% Video Capturing
v = VideoWriter('Multi_Crowd_Problem.avi'); % AVI file
v.FrameRate = 30; % optional
open(v);

while t < t_total

    % Coupling Term
```

```
    rho = rho_a + rho_b + rho_c;

    % Computing cost for the density (rho)
    speed_A = makeSpeed(rho,v_max_a,rho_max_a);
    speed_B = makeSpeed(rho,v_max_b,rho_max_b);
    discomfort_A = makeDiscomfort(rho,k_a);
    discomfort_B = makeDiscomfort(rho,k_b);
    cost_A = makeCost(speed_A,discomfort_A);
    cost_B = makeCost(speed_B,discomfort_B);

    %-- Calculate Initial Phi guess using Lower order Fast Sweep
    phi_first_order_A =
First_order_Godunov_Fast_Sweep_Algorithm("A",cost_A,h,INFINITE);
    phi_first_order_B =
First_order_Godunov_Fast_Sweep_Algorithm("B",cost_B,h,INFINITE);
    %-- Calculate The final phi using higher order scheme and above initial guess
    phi_A =
weno_fastsweep("A",cost_A,h,eta,error_eta,INFINITE,phi_first_order_A,iteration_error
);
    phi_B =
weno_fastsweep("B",cost_B,h,eta,error_eta,INFINITE,phi_first_order_B,iteration_error
);
    % Extrapolation of the phi values
    phi_A = extrapolate_phi(phi_A);
    phi_B = extrapolate_phi(phi_B);

    %--Calculate the Components of Velocity
    makeComponent("A",phi_A,speed_A);
    makeComponent("B",phi_B,speed_B);

    % Updating the delta_t with proper CFL condition (Stochastic paper)
    u_max = max([abs(x_velocity_A(:)); abs(x_velocity_B(:))]);
    v_max = max([abs(y_velocity_A(:)); abs(y_velocity_B(:))]);
    %dt = CFL / ( ( u_max/h ).^(5/3) + ( v_max/h ).^(5/3) + 1e-12 );
    dt = CFL * h / ( ( u_max ) + ( v_max ) + 1e-12 );

    % -------- Making rho_n and RHS of rho_n (Crowd A and B)-----%
    rho_n_A = rho_a;
    rho_n_B = rho_b;

    semi_discritze_form_rho_n_A = makeRHS("A",rho_n_A,t);
    semi_discritze_form_rho_n_B = makeRHS("B",rho_n_B,t);

    % Check for the negative rho_n values
    if (any(rho_n_A(:) < stop_low)|| any(rho_n_B(:) < stop_low))
        fprintf('step = %d\n',step);
        error('Negative density detected at rho_n!');
    elseif (any(rho_n_A(:) > stop_high) ||  any(rho_n_B(:) > stop_high))
        fprintf('step = %d\n',step);
        error('Density more than 10 detected at rho_n!');
```

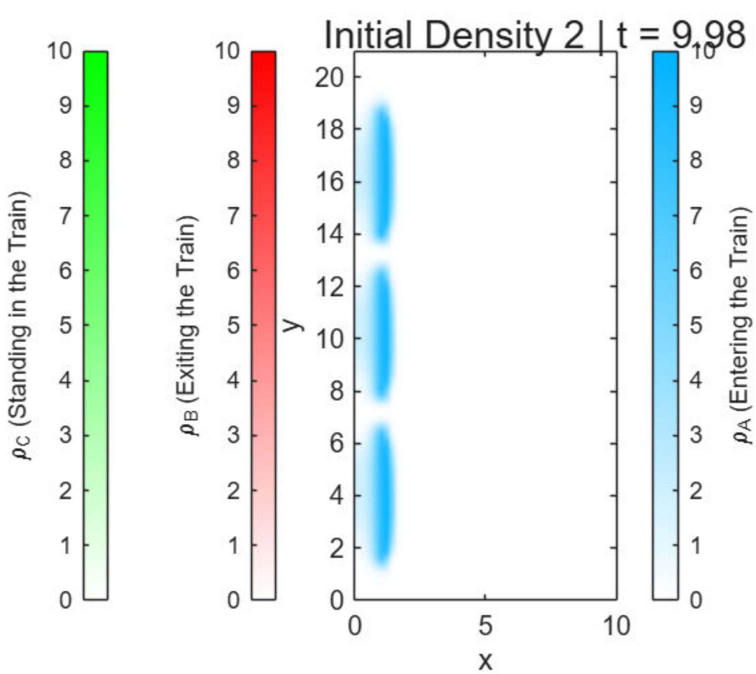
```
    end

    % Extrapolation
    rho_n_A = extrapolate_Density("A",rho_n_A);
    rho_n_B = extrapolate_Density("B",rho_n_B);


    % ---------- Making the RHO_1 and RHS of rho_1 (Crowd A and B)------ %
    for i = 4:1:Nx-3
        for j = 4:1:Ny-3
            if(~obstacle(j,i))
                rho_1_A(j,i) = rho_n_A(j,i) + dt *
semi_discritze_form_rho_n_A(j-3,i-3);
                rho_1_B(j,i) = rho_n_B(j,i) + dt *
semi_discritze_form_rho_n_B(j-3,i-3);
            else
                rho_1_A(j,i) = 0;
                rho_1_B(j,i) = 0;
            end
        end
    end


    % -- Extrapolation (My Intuition)
    rho_1_A = extrapolate_Density("A",rho_1_A);
    rho_1_B = extrapolate_Density("B",rho_1_B);

    % -- RHS of discretized form using rho_1
    semi_discritze_form_rho_1_A = makeRHS("A",rho_1_A,t);
    semi_discritze_form_rho_1_B = makeRHS("B",rho_1_B,t);

    % Check for the negative rho_1 values
    if (any(rho_n_A(:) < stop_low)|| any(rho_n_B(:) < stop_low))
        fprintf('step = %d\n',step);
        error('Negative density detected at rho_n!');
    elseif (any(rho_n_A(:) > stop_high) ||  any(rho_n_B(:) > stop_high))
        fprintf('step = %d\n',step);
        error('Density more than 10 detected at rho_n!');
    end



    %--------- Making the RHO_2 and RHS of RHO_2 ----------%
    for i = 4:1:Nx-3
        for j = 4:1:Ny-3
            if(~obstacle(j,i))
                rho_2_A(j,i) = 3/4 * rho_n_A(j,i) + 1/4 * (rho_1_A(j,i) + dt *
semi_discritze_form_rho_1_A(j-3,i-3));
                rho_2_B(j,i) = 3/4 * rho_n_B(j,i) + 1/4 * (rho_1_B(j,i) + dt *
semi_discritze_form_rho_1_B(j-3,i-3));
```

```
            else
                rho_2_A(j,i) = 0;
                rho_2_B(j,i) = 0;
            end

        end
    end


    % -- Extrapolation (My Intuition)
    rho_2_A = extrapolate_Density("A",rho_2_A);
    rho_2_B = extrapolate_Density("B",rho_2_B);

    % -- RHS of discretized form using rho_2
    semi_discritze_form_rho_2_A = makeRHS("A",rho_2_A,t);
    semi_discritze_form_rho_2_B = makeRHS("B",rho_2_B,t);

    % Check for the negative rho_2 values
    if (any(rho_n_A(:) < stop_low)|| any(rho_n_B(:) < stop_low))
        fprintf('step = %d\n',step);
        error('Negative density detected at rho_n!');
    elseif (any(rho_n_A(:) > stop_high) ||  any(rho_n_B(:) > stop_high))
        fprintf('step = %d\n',step);
        error('Density more than 10 detected at rho_n!');
    end

    %---------- Making the rho_next (Crowd A and B)----------------%

    for i = 4:1:Nx-3
        for j = 4:1:Ny-3
            if(~obstacle(j,i))
                rho_next_A(j,i) = 1/3 * rho_n_A(j,i) + 2/3 * (rho_2_A(j,i) + dt *
semi_discritze_form_rho_2_A(j-3,i-3));
                rho_next_B(j,i) = 1/3 * rho_n_B(j,i) + 2/3 * (rho_2_B(j,i) + dt *
semi_discritze_form_rho_2_B(j-3,i-3));
            else
                rho_next_A(j,i) = 0;
                rho_next_B(j,i) = 0;
            end
        end
    end

    % -- Extrapolation (My Intuition)
    rho_next_A = extrapolate_Density("A",rho_next_A);
    rho_next_B = extrapolate_Density("B",rho_next_B);

    % Substituting back the rho value
    rho_a = rho_next_A;
    rho_b = rho_next_B;
    rho = rho_next_A + rho_next_B;
```

```
 %--Check if any of the density values is equal to zero
 if (any(rho_n_A(:) < stop_low)|| any(rho_n_B(:) < stop_low))
     fprintf('step = %d\n',step);
     error('Negative density detected at rho_n!');
 elseif (any(rho_n_A(:) > stop_high) || any(rho_n_B(:) > stop_high))
     fprintf('step = %d\n',step);
     error('Density more than 10 detected at rho_n!');
 end

if mod(step,2) == 0

     figure(6);
     clf;

     % Crowd A : Dark Blue -> Bright Blue
     blueMap = [linspace(1,0,256)', linspace(1,0.7,256)', linspace(1,1,256)'];

     % Crowd B : Dark Red -> Bright Red
     redMap = [linspace(1,1,256)', linspace(1,0,256)', linspace(1,0,256)'];

     % Crowd C : White -> Green
     greenMap = [linspace(1,0,256)', linspace(1,1,256)', linspace(1,0,256)'];

     ax1 = axes;

     h1 = imagesc(x_plot, y_plot,  rho_a(4:Ny-3,4:Nx-3));

     set(gca,'YDir','normal');
     set(h1,'Interpolation','bilinear');

     axis equal tight;

     xlim([0 Lx]);
     ylim([0 Ly]);

     clim([0 10]);

     colormap(ax1, blueMap);

     cb1 = colorbar;
     cb1.Label.String = '\rho_A (Entering the Train)';

     xlabel('x');
     ylabel('y');

     ax2 = axes;

     h2 = imagesc(x_plot, y_plot, rho_b(4:Ny-3,4:Nx-3));
```

```
set(gca,'YDir','normal');
set(h2,'Interpolation','bilinear');

alphaData = rho_b(4:Ny-3,4:Nx-3) ./ 10;

set(h2, 'AlphaData', alphaData);

% Transparent background
set(ax2, 'Color', 'none');

axis equal tight;

xlim([0 Lx]);
ylim([0 Ly]);

clim([0 10]);

% Match positions
ax2.Position = ax1.Position;

% Second colormap
colormap(ax2, redMap);

% Second colorbar
cb2 = colorbar;

cb2.Label.String = '\rho_B (Exiting the Train)';

% Move second colorbar
cb2.Position(1) = cb2.Position(1) - 0.3;

% Remove duplicate ticks
ax2.XTick = [];
ax2.YTick = [];

ax4 = axes;

h3 = imagesc(x_plot, y_plot, rho_c(4:Ny-3,4:Nx-3));

set(gca,'YDir','normal');
set(h3,'Interpolation','bilinear');

alphaDataC = rho_c(4:Ny-3,4:Nx-3) ./ 10;
set(h3,'AlphaData', alphaDataC);

set(ax4,'Color','none');

axis equal tight;

xlim([0 Lx]);
```

```
        ylim([0 Ly]);

        clim([0 10]);

        ax4.Position = ax1.Position;

        colormap(ax4, greenMap);

        cb3 = colorbar;
        cb3.Label.String = '\rho_C (Standing in the Train)';

        cb3.Position(1) = cb3.Position(1) - 0.45;

        ax4.XTick = [];
        ax4.YTick = [];

        sgtitle(sprintf('Initial Density 2 | t = %.2f', t));

        drawnow;

        frame = getframe(gcf);
        writeVideo(v, frame);

    end

    % Updating the time
    t = t + dt;

    step = step + 1;
end
```



```
% Close the Video
close(v);
```

Initial Condition on the density

So there are two kinds of crowd we are dealing with. The first one being one which wants to move out of the train. Another one is the one which wants to enter the train. The one which enters the train is Crowd A and the one which exits the train is Crowd B. Given below is the picture of both the crowd initially in their respective regions.

```
function rho = makeInitialDensity(type)
    global Ny Nx;
    switch type
        % If the crowd is of the platform waiting for the train.
        case 'A'
            rho = zeros(Ny,Nx); % First make this zero
            % And now assign the density to be the equal 4.18 at the
            % location outside the train.
            rho(5:Ny-4,24:36) = 2;
            % the number 4.18 is obtained by considering that there are 300
            % people in the crowd waiting for the train.
        case 'B'
            % This is the crowd that is inside the train exiting it.
            rho = zeros(Ny,Nx);
            % I am assuming that there are equal number of people inside
            % the train with the same area.
            rho(5:Ny-4,11:17) = 2;
        case 'C'
            % This is the crowd that is inside the train standing.
            rho = zeros(Ny,Nx);
            % I am assuming that there are equal number of people inside
            % the train with the same area.
            rho(5:Ny-4,11:17) = 0;
    end
end
```

Components of Velocity

```
function makeComponent(type,phi,speed)
    global Nx Ny eta h x_velocity_A y_velocity_A x_velocity_B y_velocity_B;

    [phi_x, phi_y] = gradient(phi, h, h); % partial derivatives of phi in x and y
    mag_phi = sqrt(phi_x.^2 + phi_y.^2);
    mag_phi = max(mag_phi, eta);           % To be safe with denominator

    switch type
        case "A"
            for i = 1:1:Nx
                for j = 1:1:Ny
```

```
                    x_velocity_A(j,i) = speed(j,i) * (- phi_x(j,i))/mag_phi(j,i);
                    y_velocity_A(j,i) = speed(j,i) * (- phi_y(j,i))/mag_phi(j,i);
                end
            end
        case "B"
            for i = 1:1:Nx
                for j = 1:1:Ny
                    x_velocity_B(j,i) = speed(j,i) * (- phi_x(j,i))/mag_phi(j,i);
                    y_velocity_B(j,i) = speed(j,i) * (- phi_y(j,i))/mag_phi(j,i);
                end
            end
    end
end
```

First order Godunov Fast Sweep Algorithm (Older Version)

```
function phi = First_order_Godunov_Fast_Sweep_Algorithm(type,c,h,INFINITE)
    global Nx Ny obstacle;
    phi = INFINITE*ones(Ny, Nx);    % Initial potential field

    % Exit will depend on the type of crowd
    switch type
        case "A"
            % For crowd that is on the platform waiting or the train
            phi(4:Ny-3,4:10) = 0; % Inside the train
        case "B"
            % Fro crowd that is waiting inside the train
            phi(4:Ny-3,Nx-3) = 0; % Right Boundary
    end

    % Sweep Iterations
    for sweep = 1:20

        % Sweep 1:
        for i = 4:Nx-2
            for j = 4:Ny-2
                if ~obstacle(j, i)
                    phi = update_phi(phi, c, i, j, h);
                else
                    phi(j,i) = INFINITE;
                end
            end
        end

        % Sweep 2:
        for i = Nx-1:-1:2
            for j = 2:Ny-1
                if ~obstacle(j, i)
                    phi = update_phi(phi, c, i, j, h);
                else
```

```
                phi(j,i) = INFINITE;
            end
        end
    end

    % Sweep 3:
    for i = Nx-1:-1:2
        for j = Ny-1:-1:2
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
            else
                phi(j,i) = INFINITE;
            end
        end
    end

    % Sweep 4:
    for i = 2:Nx-1
        for j = Ny-1:-1:2
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
            else
                phi(j,i) = INFINITE;
            end
        end
    end

    % Extrapolation of the phi values

    % Bottom Edge   (3 Ghost Points)
    phi(3,:) = phi(4,:);
    phi(2,:) = phi(3,:);
    phi(1,:) = phi(2,:);

    % Top Edge      (3 Ghost Points)
    phi(Ny-3,:) = phi(Ny-2,:);
    phi(Ny-1,:) = phi(Ny-2,:);
    phi(Ny,:) = phi(Ny-1,:);

    % Right Edge    (3 Ghost Points)
    phi(:,Nx-3) = phi(:,Nx-2);
    phi(:,Nx-1) = phi(:,Nx-2);
    phi(:,Nx) = phi(:,Nx-1);

    % Left Edge     (3 Ghost Points)
    phi(:,3) = phi(:,4);
    phi(:,2) = phi(:,3);
    phi(:,1) = phi(:,2);
  end
end
```

```
% Subfunction: Update one grid point using Godunov scheme

function phi = update_phi(phi, c, i, j, dx)
    a = min(phi(j, i-1), phi(j, i+1)); % x-direction neighbors
    b = min(phi(j-1, i), phi(j+1, i)); % y-direction neighbors
    c_i = c(j, i); % Local inverse speed
    if abs(a - b) >= c_i * dx
        phi(j, i) = min(a, b) + c_i * dx;
        else
            inside = 2 * (c_i * dx)^2 - (a - b)^2;
        if inside >= 0
            phi(j, i) = (a + b + sqrt(inside)) / 2;
        end
    end
end
```

WENO FAST SWEEP

```
function phi =
weno_fastsweep(type,c,h,eta,error_eta,INFINITE,initial_guess,iteration_error)

    global Nx Ny obstacle;

    % Phi is initialized with infinity
    phi = initial_guess;
    phi_old = INFINITE * ones(size(phi));

    % Iterations
    iterations = 0;
    loop_safety = 0;

    % Iterations
    while (sum(abs(phi(:)-phi_old(:))) > error_eta && loop_safety<1000)

        %fprintf("Iteration = %d ",iterations);
        err = sum(abs(phi(:)-phi_old(:)));
        %fprintf("Error = %0.10f\n",err);

        phi_old = phi;
        iterations = iterations + 1;
        iteration_error(iterations) = err;
        loop_safety = loop_safety + 1;

        for sweep = 1:4
            switch sweep
                case 1
                    ix = 4:1:Nx-3; jy = 4:1:Ny-3;
                case 2
```

```
                ix = 4:1:Nx-3; jy = Ny-3:-1:4;
            case 3
                ix = Nx-3:-1:4; jy = 4:1:Ny-3;
            case 4
                ix = Nx-3:-1:4; jy = Ny-3:-1:4;
        end

        for i = ix
            for j = jy
                %if the point is outside the obstacle
                if(~obstacle(j,i))
                    %------------Calcualting the phix_min---------_%

                    r_back = (eta + (phi(j,i) - 2*phi(j,i-1) + phi(j,i-2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);
                    r_front = (eta + (phi(j,i) - 2*phi(j,i+1) + phi(j,i+2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);

                    % Calculate the w
                    w_front = 1/(1+2*(r_front^2));
                    w_back = 1/(1+2*(r_back^2));

                    % Dell phi by Dell x plus and minus
                    phix_minus = (1-w_back)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_back*((3*phi(j,i) - 4*phi(j,i-1) + phi(j,i-2))/(2*h));
                    phix_plus = (1-w_front)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_front*((-3*phi(j,i)+4*phi(j,i+1)-phi(j,i+2))/(2*h));


                    phix_min = min((phi(j,i) - h*phix_minus),(phi(j,i) +
h*phix_plus));

                    %-----------calculating the phiy_min---------%

                    r_back = (eta + (phi(j,i) - 2*phi(j-1,i) + phi(j-2,i))^2) /
(eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);
                    r_front = (eta + (phi(j,i) - 2*phi(j+1,i) +
phi(j+2,i))^2) / (eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);

                    % Calculate the w
                    w_front = 1 / (1 + 2*(r_front^2));
                    w_back = 1 / (1 + 2*(r_back^2));

                    % Dell phi by Dell x plus and minus
                    phiy_minus = (1-w_back) * ((phi(j+1,i)-phi(j-1,i))/(2*h)) +
w_back * ((3*phi(j,i) - 4*phi(j-1,i) + phi(j-2,i))/(2*h));
                    phiy_plus = (1-w_front) * ((phi(j+1,i) - phi(j-1,i))/(2*h))
+ w_front * ((-3*phi(j,i) + 4*phi(j+1,i) - phi(j+2,i))/(2*h));
```

```
                        phiy_min = min((phi(j,i) - h*phiy_minus), (phi(j,i) +
h*phiy_plus));

                        if abs(phix_min - phiy_min) >= c(j,i)*h
                            phi(j,i) = min(phiy_min,phix_min) + c(j,i)*h;
                        else
                            phi(j,i) = ((phix_min + phiy_min) + (2*(c(j,i)*h)^2 -
(phix_min - phiy_min)^2)^0.5)/2;
                        end

                        % Enforce the value of phi to be zero at exits depending on
the crowd
                        switch type
                            case "B"
                                % Right Boundary for the crowd inside the
                                % train
                                if(i == Nx-3 && (j >= 4 && j <= Ny-3))
                                    phi(j,i) = 0;
                                end
                            case "A"
                                % Top Boundary for the crowd on the
                                % platform
                                if((j >= 4 && j < Ny-3) && (i >= 4 && i <= 10))
                                    phi(j,i) = 0;
                                end
                        end

                    % If the point is inside the obstacle
                    else
                        phi(j,i) = INFINITE;
                    end
                end
            end
        end
    end
end
```

Extrapolate Phi

```
function phi = extrapolate_phi(phi)
    global Ny Nx;
    % Bottom Edge   (3 Ghost Points)
    phi(3,:) = phi(4,:);
    phi(2,:) = phi(3,:);
    phi(1,:) = phi(2,:);
    % Top Edge      (3 Ghost Points)
    phi(Ny-2,:) = phi(Ny-3,:);
    phi(Ny-1,:) = phi(Ny-2,:);
    phi(Ny,:) = phi(Ny-1,:);
```

```
    % Right Edge    (3 Ghost Points)
    phi(:,Nx-2) = phi(:,Nx-3);
    phi(:,Nx-1) = phi(:,Nx-2);
    phi(:,Nx) = phi(:,Nx-1);
    % Left Edge     (3 Ghost Points)
    phi(:,3) = phi(:,4);
    phi(:,2) = phi(:,3);
    phi(:,1) = phi(:,2);
end
```

RHS of Semi Discretize Form

```
function semi_discrite_form =  makeRHS(type,rho,t)
    global Nx Ny x_velocity_A y_velocity_A x_velocity_B y_velocity_B h obstacle;
    semi_discrite_form = zeros(Ny-6,Nx-6);

    % Flux for the given rho distribution (Will depend on the type of crowd)
    switch type
        case "A"
            x_velocity = x_velocity_A;
            y_velocity = y_velocity_A;
        case "B"
            x_velocity = x_velocity_B;
            y_velocity = y_velocity_B;
    end
    flux = makeFlux(rho, x_velocity, y_velocity);
    flux_x = squeeze(flux(1,:,:));
    flux_y = squeeze(flux(2,:,:));

    % Calculating the maximum speed (multiplying by 1.5 as suggested)
    alpha_x = max(abs(x_velocity(:))) * 1.5;
    alpha_y = max(abs(y_velocity(:))) * 1.5;


    % -- Calculating the value of f+ and f- at {i+2,i+1,i,i-1,i-2,i-3} -- %
    fx_negative_whole = zeros(Ny,Nx);
    fx_positive_whole = zeros(Ny,Nx);
    fy_negative_whole = zeros(Ny,Nx);
    fy_positive_whole = zeros(Ny,Nx);

    for j = 1:1:Ny
        for i=1:1:Nx
            fx_positive_whole(j,i) = 1/2 * (flux_x(j,i) + alpha_x*rho(j,i));
            fx_negative_whole(j,i) = 1/2 * (flux_x(j,i) - alpha_x*rho(j,i));
            fy_positive_whole(j,i) = 1/2 * (flux_y(j,i) + alpha_y*rho(j,i));
            fy_negative_whole(j,i) = 1/2 * (flux_y(j,i) - alpha_y*rho(j,i));
        end
    end

    % -- Computing the RHS of Semi Discrete form.
```

```
    for i = 4:1:Nx-3
        for j = 4:1:Ny-3
            % compute only at the points where obstacle doesn't exist
            if(~obstacle(j,i))
                %--X--%

                % fx{+}_[i+1/2,j] - Given in the paper
                fx_plus_x_half_forward = stencil( ...
                    fx_positive_whole(j,i-2), ...
                    fx_positive_whole(j,i-1), ...
                    fx_positive_whole(j,i), ...
                    fx_positive_whole(j,i+1), ...
                    fx_positive_whole(j,i+2));

                % fx{+}_[i-1/2,j] - By substituting i -> (i-1) in fx{+}_[i+1/2,j]
                fx_plus_x_half_backward = stencil( ...
                    fx_positive_whole(j,i-3), ...
                    fx_positive_whole(j,i-2), ...
                    fx_positive_whole(j,i-1), ...
                    fx_positive_whole(j,i), ...
                    fx_positive_whole(j,i+1));

                % fx{-}_[i+1/2,j] ( mirror of fx{+}_[i+1/2,j] )
                fx_minus_x_half_forward = stencil( ...
                    fx_negative_whole(j,i+3), ...
                    fx_negative_whole(j,i+2), ...
                    fx_negative_whole(j,i+1), ...
                    fx_negative_whole(j,i), ...
                    fx_negative_whole(j,i-1));

                % fx{-}_[i-1/2,j] ( mirror of fx{+}_[i-1/2,j] )
                fx_minus_x_half_backward = stencil( ...
                    fx_negative_whole(j,i+2), ...
                    fx_negative_whole(j,i+1), ...
                    fx_negative_whole(j,i), ...
                    fx_negative_whole(j,i-1), ...
                    fx_negative_whole(j,i-2));

                %--Y--%

                % fy{+}_[i,j+1/2]
                fy_plus_y_half_forward = stencil( ...
                    fy_positive_whole(j-2,i), ...
                    fy_positive_whole(j-1,i), ...
                    fy_positive_whole(j,i), ...
                    fy_positive_whole(j+1,i), ...
                    fy_positive_whole(j+2,i));

                % fy{+}_[i,j-1/2]
                fy_plus_y_half_backward = stencil( ...
```

```
                    fy_positive_whole(j-3,i), ...
                    fy_positive_whole(j-2,i), ...
                    fy_positive_whole(j-1,i), ...
                    fy_positive_whole(j,i), ...
                    fy_positive_whole(j+1,i));

                % fy{-}_[i,j+1/2]
                fy_minus_y_half_forward = stencil( ...
                    fy_negative_whole(j+3,i), ...
                    fy_negative_whole(j+2,i), ...
                    fy_negative_whole(j+1,i), ...
                    fy_negative_whole(j,i), ...
                    fy_negative_whole(j-1,i));

                % fy{-}_[i,j-1/2]
                fy_minus_y_half_backward = stencil( ...
                    fy_negative_whole(j+2,i), ...
                    fy_negative_whole(j+1,i), ...
                    fy_negative_whole(j,i), ...
                    fy_negative_whole(j-1,i), ...
                    fy_negative_whole(j-2,i));

                fx_positive_half = fx_plus_x_half_forward + fx_minus_x_half_forward;
                fx_negative_half = fx_plus_x_half_backward +
fx_minus_x_half_backward;
                fy_positive_half = fy_plus_y_half_forward + fy_minus_y_half_forward;
                fy_negative_half = fy_plus_y_half_backward +
fy_minus_y_half_backward;


                %--Boundary Condition -- %

                if(j == Ny-3)
                    fy_positive_half = 0;
                elseif(j == 4)
                    fy_negative_half = 0;
                end

                if(i == Nx-3)
                    if(type == 'A')
                        fx_positive_half = 0;
                    end
                    % For the crowd of type B, this is a wall
                elseif(i == 4)
                    fx_negative_half = 0;
                end

                % BOUNDARY OF OBSTACLE
                % Obstacle 1
                if(j >= 1 && j <= 18)
```

```
        % Boundary of Obstacle 1
        if(i == 18)
            fx_positive_half = 0;
        elseif(i == 21)
            fx_negative_half = 0;
        end
    elseif (j == 19 && (i >= 19 && i <= 20))
        % Corner of Obstacle 1
        fy_negative_half = 0;
        fx_negative_half = 0;
    end


    % Obstacle 2
    if(j >= 34 && j <= 48)
        % Boundary of Obstacle 2
        if(i == 18)
            fx_positive_half = 0;
        elseif(i == 21)
            fx_negative_half = 0;
        end
    elseif (j == 33 && (i >= 19 && i <= 20))
        % Corner of Obstacle 2
        fy_positive_half = 0;
        fx_positive_half = 0;
    elseif(j == 49 && (i >= 19 && i <= 20))
        % Corner of Obstacle 2
        fy_negative_half = 0;
        fx_negative_half = 0;
    end

    % Obstacle 3
    if(j >= 64 && j <= 78)
        % Boundary of Obstacle 2
        if(i == 18)
            fx_positive_half = 0;
        elseif(i == 21)
            fx_negative_half = 0;
        end
    elseif (j == 63 && (i >= 19 && i <= 20))
        % Corner of Obstacle 2
        fy_positive_half = 0;
        fx_positive_half = 0;
    elseif(j == 79 && (i >= 19 && i <= 20))
        % Corner of Obstacle 2
        fy_negative_half = 0;
        fx_negative_half = 0;
    end

    % Obstacle 4
```

```
                if(j >= 94 && j <= Ny)
                    % Boundary of Obstacle 4
                    if(i == 18)
                        fx_positive_half = 0;
                    elseif(i == 21)
                        fx_negative_half = 0;
                    end
                elseif(j == 93 && (i <= 20 && i >= 19))
                    % Corner of obstacle 4
                    fx_positive_half = 0;
                    fy_positive_half = 0;
                end

                % Computing RHS of Semi Discretized form
                semi_discrite_form(j-3,i-3) = - 1/h * (fx_positive_half -
fx_negative_half) - 1/h * (fy_positive_half - fy_negative_half);

            else

            end
        end
    end
end
```

WENO Advection Stencil

```
% To compute f half. Order of input matters
function f = stencil(f1,f2,f3,f4,f5)

    eta = 10^-6;

    % Compute S values
    s_1 = ((1/3) * f1) + ((-7/6) * f2) + ((11/6) * f3);
    s_2 = ((-1/6) * f2) + ((5/6) * f3) + ((1/3) * f4);
    s_3 = ((1/3) * f3) + ((5/6) * f4) + ((-1/6) * f5);

    % Compute Beta
    beta_1 = 13/12 * (f1 - 2*f2 + f3)^2 + 1/4 * (f1 - 4*f2 + 3*f3)^2;
    beta_2 = 13/12 * (f2 - 2*f3 + f4)^2 + 1/4 * (f2 - f4)^2;
    beta_3 = 13/12 * (f3 - 2*f4 + f5)^2 + 1/4 * (3*f3 - 4*f4 + f5)^2;

    % Defining Gamma
    gamma_1 = 1/10;
    gamma_2 = 3/5;
    gamma_3 = 3/10;

    % Computing weights bar
    w_1_dash = ( gamma_1 ) ./ ((eta + beta_1)^2);
    w_2_dash = ( gamma_2 ) ./ ((eta + beta_2)^2);
    w_3_dash = ( gamma_3 ) ./ ((eta + beta_3)^2);
```

```
    % Computing Weights
    w_1 = ( w_1_dash ) / (w_1_dash + w_2_dash + w_3_dash);
    w_2 = ( w_2_dash ) / (w_1_dash + w_2_dash + w_3_dash);
    w_3 = ( w_3_dash ) / (w_1_dash + w_2_dash + w_3_dash);

    % Computing Flux
    f = w_1 * s_1 + w_2 * s_2 + w_3 * s_3;
end
```

Density Extrapolation

```
function rho = extrapolate_Density(type,rho)
    global Ny Nx;
    % Left Edge     (3 Ghost Points)
    rho(:,3) = rho(:,4);
    rho(:,2) = rho(:,3);
    rho(:,1) = rho(:,2);
    % Bottom Edge   (3 Ghost Points)
    rho(3,:) = rho(4,:);
    rho(2,:) = rho(3,:);
    rho(1,:) = rho(2,:);
    % Top Edge      (3 Ghost Points)
    rho(Ny-2,:) = rho(Ny-3,:);
    rho(Ny-1,:) = rho(Ny-2,:);
    rho(Ny,:) = rho(Ny-1,:);
    switch type
        case "A"
            % Right Edge    (3 Ghost Points)
            rho(:,Nx-2) = rho(:,Nx-3);
            rho(:,Nx-1) = rho(:,Nx-2);
            rho(:,Nx) = rho(:,Nx-1);
        case "B"
            % Right Edge    (3 Ghost Points)
            rho(4:Ny-3,Nx-3) = 0; % Exit 2
            rho(:,Nx-2) = rho(:,Nx-3);
            rho(:,Nx-1) = rho(:,Nx-2);
            rho(:,Nx) = rho(:,Nx-1);
    end
end
```